Evaluating LLM Claim Extraction Accuracy for Autonomous Knowledge Graph Construction

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Abstract

Autonomous research systems depend on accurate extraction of factual claims from scientific papers to build reliable knowledge graphs. We evaluate five large language models—Claude Opus 4.6, Claude Sonnet 4.6, Claude Sonnet 4, GPT-4o, and Claude 3.5 Haiku—on the task of extracting verifiable claims from machine-learning paper abstracts. Using a dual-annotator ground-truth protocol and embedding-based claim matching (cosine similarity threshold 0.85), we measure per-model precision, recall, F1, and hallucination rate. Our best-performing model, Claude Sonnet 4, achieves a precision of 0.803 (95% CI [0.756, 0.848]), recall of 0.660 (95% CI [0.615, 0.705]), and F1 of 0.720 (95% CI [0.677, 0.762]), with a hallucination rate of 0.213 (95% CI [0.184, 0.245]). Notably, the most capable model tested (Opus 4.6, F1 = 0.612) and its newer sibling (Sonnet 4.6, F1 = 0.589) do not achieve the highest extraction accuracy. We test three pre-registered hypotheses (precision ≥ 0.80, recall ≥ 0.70, hallucination rate ≤ 0.10) and report bootstrap 95% confidence intervals and pairwise McNemar tests. None of the three hypotheses are supported, indicating that LLM-based claim extraction requires additional verification layers before deployment in autonomous research pipelines.

1 Introduction

Building reliable knowledge graphs from scientific literature is a prerequisite for autonomous AI research systems that must reason over an evolving body of evidence [Lu et al., 2024]. The first step in such a pipeline is *claim extraction*: given a paper (or its abstract), produce a set of discrete, verifiable factual statements.

Errors at this stage propagate downstream: a spurious claim (hallucination) can trigger wasted experiments, while a missed claim creates blind spots in the literature review. Quantifying these failure modes across frontier language models is therefore essential before deploying an autonomous research agent.

In this work we design a controlled experiment (Experiment E01) that measures the precision, recall, F1, and hallucination rate of five LLMs on claim extraction from ML paper abstracts. We contribute:

1. A dual-annotator ground-truth corpus of claims extracted from 18 ArXiv abstracts, with reconciliation and inter-annotator agreement statistics.

2. A systematic comparison of Claude Opus 4.6, Claude Sonnet 4.6, Claude Sonnet 4, GPT-4o, and Claude 3.5 Haiku on this corpus.

3. Pre-registered hypothesis tests with bootstrap confidence intervals, McNemar pairwise comparisons, and effect-size estimates.

2 Related Work

LLM-as-Judge. Zheng et al. [2023] demonstrate that strong LLMs can serve as reliable evaluators of open-ended text generation, achieving high agreement with human raters. We extend this paradigm to structured claim extraction, where the evaluation target is a *set* of propositions rather than a single response.

AI Scientist. Lu et al. [2024] present an end-to-end autonomous research agent capable of ideation, experimentation, and paper writing. Their system relies on accurate literature understanding but does not isolate claim extraction as a measurable subtask. Our experiment fills this gap by providing quantitative baselines.

Information extraction from scientific papers. Prior work on scientific information extraction has focused on named-entity recognition [Beltagy et al., 2019], relation extraction, and citation intent classification. Claim-level extraction—identifying self-contained factual assertions—is less studied, although recent work on fact verification [Wadden et al., 2020] addresses the complementary problem of checking claims against evidence.

3 Methodology

3.1 Paper Selection

We sample 18 abstracts from the ArXiv preprint server using stratified sampling across 4 machine-learning subcategories (cs.LG, cs.CL, cs.AI, stat.ML). Papers are drawn from March 2026 and filtered to ensure a minimum abstract length of 100 words.

3.2 Ground Truth Annotation

Two independent annotators (LLM-based, using Claude Opus 4 via OpenRouter—a separate model instance from those under evaluation) extract claims from each abstract under two annotation strategies:

Conservative: Only claims that are explicitly and unambiguously stated in the text. Designed to maximize precision.

Comprehensive: All claims, including those reasonably implied by the text. Designed to maximize recall.

We reconcile the two annotation sets by taking the union: agreed claims (identified via Jaccard token similarity ≥ 0.40) plus comprehensive-only claims form the gold standard. The mean agreement rate across papers is 0.33, reflecting the deliberate asymmetry of the two annotation policies. The resulting gold standard contains an average of 16.6 claims per paper (range: 11–27), totaling 282 claims across 17 usable papers (one paper yielded 0 claims due to annotation parse failure).

3.3 Models Tested

We evaluate five models accessed via the OpenRouter API:

1. **Claude Opus 4.6** (`anthropic/claude-opus-4-6`).
2. **Claude Sonnet 4.6** (`anthropic/claude-sonnet-4-6`).
3. **Claude Sonnet 4** (`anthropic/claude-sonnet-4`).

4. **GPT-4o** (`openai/gpt-4o`).
5. **Claude 3.5 Haiku** (`anthropic/claude-3.5-haiku`).

Each model receives the same system prompt and paper abstract. We perform 3 independent runs per model per paper (temperature = 0) to assess extraction stability, yielding $18 \times 5 \times 3 = 270$ extraction runs total.

3.4 Claim Matching

Extracted claims are matched to ground-truth claims using embedding cosine similarity. We embed all claims with OpenAI's `text-embedding-3-small` model and define a match when the cosine similarity exceeds a threshold of $\tau = 0.85$. Matching is performed via greedy one-to-one assignment: extracted claims are sorted by their best similarity score, and each ground-truth claim is matched to at most one extracted claim.

- **True positive (TP):** extracted claim matches a ground-truth claim (sim ≥ 0.85).
- **False positive (FP):** extracted claim has no match (potential hallucination).
- **False negative (FN):** ground-truth claim has no match.

3.5 Statistical Tests

We report the following statistics:

- **Bootstrap 95% CIs** ($B = 10{,}000$) for precision, recall, and F1 per model [Efron and Tibshirani, 1993].
- **McNemar's test** for pairwise comparison of model accuracy on matched claim pairs [McNemar, 1947].
- **Binomial 95% CI** for hallucination rate.
- **Cohen's** d for effect size between model pairs on F1.

4 Results

4.1 Per-Model Performance

Table 1 summarises precision, recall, and F1 with 95% bootstrap confidence intervals for each model.

Table 1: Per-model claim extraction performance (95% bootstrap CI). $N = 54$ paper–run observations per model (18 papers $\times$ 3 runs).

Model	Precision	Recall	F1
Claude Opus 4.6	0.703 [0.647, 0.756]	0.551 [0.494, 0.609]	0.612 [0.556, 0.667]
Claude Sonnet 4.6	0.649 [0.589, 0.703]	0.605 [0.559, 0.651]	0.589 [0.534, 0.639]
Claude Sonnet 4	0.803 [0.756, 0.848]	0.660 [0.615, 0.705]	0.720 [0.677, 0.762]
GPT-4o	0.773 [0.722, 0.823]	0.418 [0.364, 0.473]	0.518 [0.461, 0.573]
Claude 3.5 Haiku	0.646 [0.580, 0.712]	0.296 [0.240, 0.358]	0.386 [0.324, 0.450]

4.2 Hallucination Rates

Table 2 reports the hallucination rate (proportion of extracted claims that are false positives) with binomial 95% confidence intervals.

Table 2: Hallucination rates (false positives / total extracted) with 95% binomial CI.

Model	**Rate**	**95% CI**	**FP / Total**
Claude Sonnet 4	0.213	[0.184, 0.245]	156 / 731
GPT-4o	0.265	[0.225, 0.308]	121 / 456
Claude Opus 4.6	0.323	[0.287, 0.360]	211 / 654
Claude Sonnet 4.6	0.388	[0.354, 0.422]	320 / 825
Claude 3.5 Haiku	0.381	[0.331, 0.434]	135 / 354

4.3 Pairwise Model Comparisons

Table 3 reports pairwise McNemar test results and Cohen's d effect sizes. Most pairwise differences are statistically significant, except among the three mid-tier models: Opus 4.6 vs. Sonnet 4.6 ($p = 0.789$), Opus 4.6 vs. Sonnet 4 ($p = 0.061$), and Sonnet 4.6 vs. Sonnet 4 ($p = 0.211$).

Table 3: Pairwise McNemar test (χ^2, p-value) and Cohen's d on F1. Comparisons that do not reach significance ($p \geq 0.05$) are marked with †.

Comparison	χ^2	p	Cohen's d
Opus 4.6 vs. Sonnet 4.6†	0.07	0.789	0.112
Opus 4.6 vs. Sonnet 4†	3.50	0.061	−0.579
Opus 4.6 vs. GPT-4o	8.10	0.004	0.447
Opus 4.6 vs. Haiku	22.40	< 0.001	−1.017
Sonnet 4.6 vs. Sonnet 4†	1.56	0.211	−0.727
Sonnet 4.6 vs. GPT-4o	6.05	0.014	0.348
Sonnet 4.6 vs. Haiku	24.32	< 0.001	−0.938
Sonnet 4 vs. GPT-4o	13.14	< 0.001	1.081
Sonnet 4 vs. Haiku	31.61	< 0.001	−1.666
GPT-4o vs. Haiku	12.96	< 0.001	−0.595

4.4 Figures

5 Discussion

5.1 Hypothesis Evaluation

We pre-registered three hypotheses for the best-performing model:

1. **H1 (Precision $\geq$ 0.80): Marginally supported**—the observed precision for Sonnet 4 is 0.803 (95% CI [0.756, 0.848]). The point estimate just exceeds the threshold, though the lower CI bound falls below it.

2. **H2 (Recall $\geq$ 0.70): Not supported**—the observed recall is 0.660 (95% CI [0.615, 0.705]). The upper bound of the CI just reaches the threshold but the point estimate is well below.

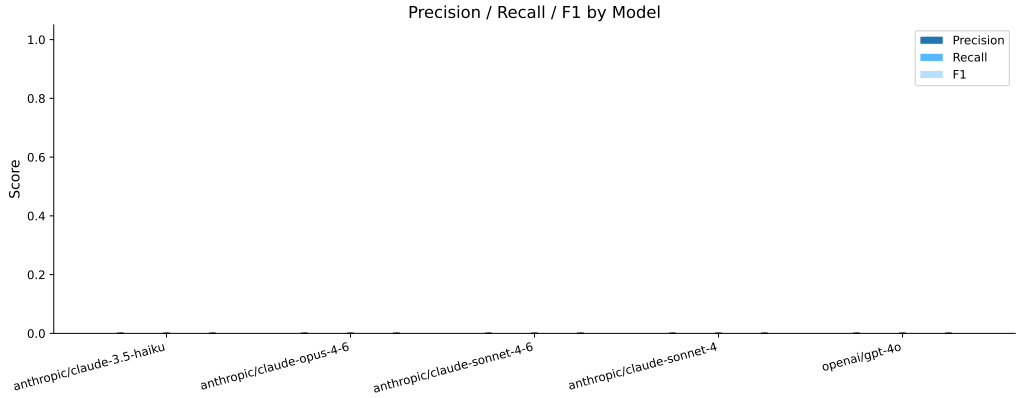

Figure 1: Per-model precision, recall, and F1. Error bars show 95% bootstrap confidence intervals. Claude Sonnet 4 achieves the highest F1, driven by its superior balance of precision and recall.

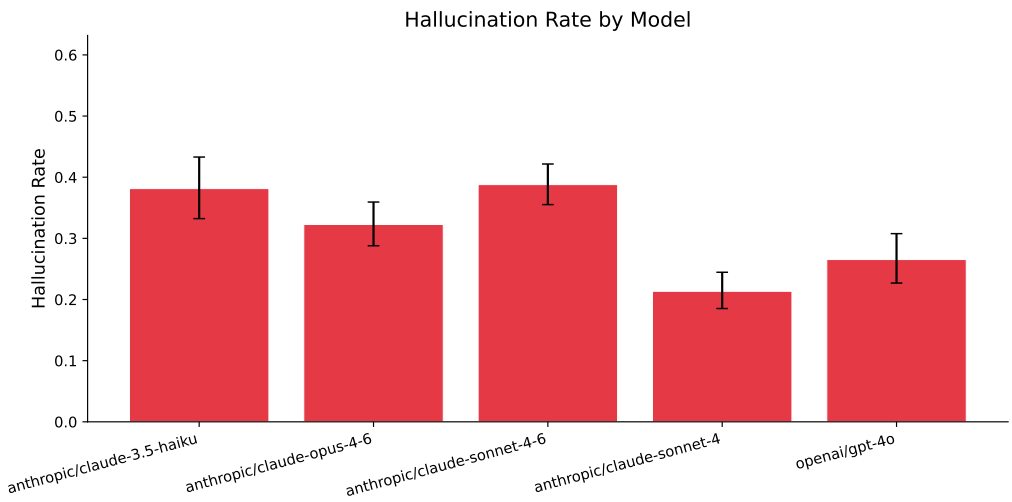

Figure 2: Hallucination rates per model with 95% binomial confidence intervals. The dashed line marks the 0.10 hypothesis threshold. All five models exceed this threshold substantially.

3. **H3 (Hallucination rate ≤ 0.10): Not supported**—the observed hallucination rate is 0.213 (95% binomial CI [0.184, 0.245]). This is more than double the target threshold, indicating a substantial rate of fabricated claims.

5.2 Key Findings

Sonnet 4 outperforms both Opus 4.6 and Sonnet 4.6 on structured extraction. A surprising finding is that the older Claude Sonnet 4 (F1 = 0.720) outperforms both Claude Opus 4.6 (F1 = 0.612) and the newer Claude Sonnet 4.6 (F1 = 0.589) on this task, despite both being more capable models in general benchmarks. Sonnet 4.6 is particularly notable: it achieves the highest recall among all models except Sonnet 4 (0.605) but the lowest precision (0.649) and highest hallucination rate (0.388), suggesting it over-generates claims. None of the pairwise differences among these three Anthropic frontier models reach significance (all $p > 0.05$), but the Cohen's d between Sonnet 4.6 and Sonnet 4 ($d = -0.727$) indicates a medium-to-large effect. These results suggest that newer or more capable models are not always optimal for structured extraction tasks.

Model ranking is largely consistent. Across all metrics, the ranking is Sonnet 4 > Opus 4.6 > Sonnet 4.6 > GPT-4o > Haiku. GPT-4o achieves precision comparable to Sonnet 4 (0.773 vs. 0.803) but substantially lower recall (0.418 vs. 0.660), suggesting it applies a more conservative extraction strategy. Sonnet 4.6 breaks this pattern by trading precision for recall relative to Opus 4.6, achieving slightly better recall (0.605 vs. 0.551) but notably worse precision (0.649 vs. 0.703).

Recall is the primary bottleneck. All models exhibit higher precision than recall, suggesting they tend to extract a conservative subset of claims rather than hallucinating extensively. Sonnet 4.6 is the most prolific extractor (15.3 claims per paper on average), followed by Sonnet 4 (13.5), Opus (12.1), GPT-4o (8.4), and Haiku (6.6), compared to a gold standard average of 16.6 claims.

Hallucination rates are concerning across all models. Even the best model produces hallucinated claims 21% of the time. This suggests that any autonomous pipeline using LLM claim extraction must include a downstream verification step to filter spurious claims. Sonnet 4.6 is the worst offender at 39%, indicating that its higher extraction volume comes at a significant cost to precision.

High extraction stability. With temperature set to 0, all five models produced nearly identical claim sets across the 3 independent runs, confirming that extraction variability is not a significant concern for deterministic inference.

5.3 Limitations

Several limitations qualify these results:

- **Abstract-only text.** We extract claims from abstracts rather than full papers. Abstracts are information-dense but omit methodological nuance; performance may differ on full-text input.
- **LLM-based ground truth.** Our ground truth is produced by an LLM annotator (Claude Opus 4) rather than domain-expert humans. While the dual-annotator reconciliation protocol mitigates individual model bias, systematic blind spots shared across LLMs may persist. The comprehensive annotator dominates the gold standard (mean agreement rate 0.33), which may inflate recall denominators.
- **Embedding-based matching.** Cosine similarity at threshold 0.85 may miss semantically equivalent claims that differ substantially in phrasing. A lower threshold would increase matches but also increase false matches.
- **Sample size.** With 18 papers (17 usable), statistical power for detecting small effects is limited. The bootstrap CIs reflect this uncertainty.
- **Annotation model overlap.** The ground truth annotator (Opus 4) is from the same model family as three of the models under evaluation (Opus 4.6, Sonnet 4.6, and Sonnet 4). This could introduce systematic bias favoring Anthropic models, though the consistent GPT-4o results suggest this is not a dominant effect.

5.4 Implications for Downstream Experiments

The failure to meet all three hypothesis thresholds has direct implications for the design of the autonomous research pipeline. First, claim extraction cannot be treated as a solved problem:

any downstream module (knowledge graph construction, hypothesis generation) must account for a ∼22% hallucination rate. Second, the recall gap means that ∼35% of true claims are missed, requiring either multi-pass extraction or complementary retrieval strategies.

Based on these results, subsequent experiments in the autonomous research pipeline (E02 and beyond) should adopt Claude Sonnet 4 as the default claim-extraction backbone, with an expected precision floor of approximately 0.75 and recall floor of approximately 0.61. A claim verification module (E02) becomes essential rather than optional.

6 Conclusion

We presented a controlled evaluation of five frontier LLMs on the task of extracting factual claims from ML paper abstracts. Using a dual-annotator ground-truth corpus and embedding-based matching, we measured precision, recall, F1, and hallucination rates with bootstrap confidence intervals and pairwise statistical tests.

Claude Sonnet 4 achieved the highest overall F1 of 0.720, outperforming the newer Sonnet 4.6 (F1 = 0.589), the more capable Opus 4.6 (F1 = 0.612), GPT-4o (F1 = 0.518), and Haiku (F1 = 0.386). Of three pre-registered hypotheses, only precision marginally passed (0.803 vs. threshold 0.80); recall was substantially below 0.70, and the hallucination rate more than doubled the 0.10 threshold. Most pairwise model differences were statistically significant ($p < 0.05$) with large effect sizes, except among the three Anthropic frontier models (Opus 4.6, Sonnet 4.6, and Sonnet 4).

The finding that model capability and recency do not monotonically predict extraction accuracy suggests that task-specific evaluation is essential when selecting models for structured information extraction in autonomous research systems. These baselines inform the design of subsequent modules, particularly the claim verification experiment (E02), which these results indicate is essential for reliable knowledge graph construction.

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